

Computer Networks — Internet Governance & Control

GATE-Level Notes

1. Internet Governance: Basic Idea

The Internet is a **network of networks** consisting of many independently operated networks. Therefore, there is **no single entity that controls every aspect of the Internet**.

Different stakeholders participate in Internet governance, including:

- Governments
- Technical organizations
- Network operators
- Businesses
- Researchers
- Civil society and users

This is called a **multi-stakeholder setting**.

Key Point

Internet Governance ≠ control by one organization.

Different organizations have roles at different layers.

2. Multi-Stakeholder Model

Different stakeholders may have **different or conflicting interests**.

Dave Clark and colleagues described this interaction as a "**tussle**".

Tussle

Tussle = interaction/conflict among different stakeholders trying to protect their respective interests.

The lecture emphasizes that such interaction plays an important role in the evolution of the Internet.

GATE Keyword

Multiple stakeholders



Different interests



Tussle



Internet governance/evolution

3. Three Layers of Internet Governance ★ ★ ★

This is the **most important part for GATE**.

Internet governance can be viewed in **three major dimensions**:

3. CONTENT	
What information is available?	
Who can access it?	
2. NAMES & NUMBERS	
Domain names + IP addresses	

1. INFRASTRUCTURE / TECHNICAL	
Hardware + protocols + formats	

Easy Memory Trick

T → N → C

Technical → Names & Numbers → Content

4. Layer 1 — Infrastructure / Technical Standards

This layer deals mainly with the **technical operation of the Internet**.

It includes standards related to:

- Hardware
- Software
- Links
- Switches
- Routers
- Protocols
- Message formats
- Data formats

Examples

Routers

Switches

Protocols

Message Formats

Data Formats



Technical Standards

Main Goal

Interoperability — different systems should be able to work together.

5. Why Are Standards Needed? ★

Without common standards, different systems may not **interoperate**.

Example:

If different networks used incompatible communication standards:

Network A



Network B

Communication would be difficult/impossible.

With common standards:

Network A — Standard — Network B



Interoperability

The lecture uses electrical systems and railroad gauges to illustrate that standardization allows independent systems to interconnect and interoperate.

GATE Line

Standardization enables interoperability among independently operated systems.

6. IETF ★ ★ ★

Full Form

IETF = Internet Engineering Task Force

It is an open international community involving:

- Network designers

- Operators
- Vendors
- Researchers

Its focus is the **evolution of Internet architecture and smooth operation of the Internet.**

IETF Role

IETF develops/standardizes technical specifications for Internet protocols.

Examples mentioned in the lecture:

- HTTP
- TCP
- IP

7. RFC ★ ★

RFC = Request for Comments

RFCs define/document many of the **technical standards and specifications** used in the Internet.

IETF



Technical standardization



RFCs



Internet protocols/specifications

GATE Trap

✗ RFC = only a comment/question document

☑ RFCs are documents used to specify/document Internet standards and protocols.

8. 3GPP

Full Form

3GPP = 3rd Generation Partnership Project

It develops technical standards for:

- 3G
- 4G
- 5G

mobile cellular systems.

The lecture describes it as an alliance of standards organizations working on areas such as wireless access networks, core networks and terminals, and overall system/service aspects.

Remember

3GPP → Cellular / Mobile standards

3G, 4G, 5G

9. IEEE

The lecture mentions IEEE standards for:

IEEE 802.3

Associated with **Ethernet** standards.

IEEE 802.11

Associated with **Wi-Fi / wireless LAN** standards.

GATE Memory

802.3 → Ethernet

802.11 → Wi-Fi

10. ITU

ITU = International Telecommunication Union

It is an international telecommunications standards organization and has been a specialized agency of the United Nations since 1949, according to the lecture.

11. Layer 2 — Internet Names & Numbers ★ ★ ★

This layer deals with:

- **Domain names**
- **IP addresses**
- Assignment/coordination of names and addresses
- Mapping between names and IP addresses

Example:

Domain Name

gaia.cs.umass.edu

↓

DNS

↓

IP Address

128.119.40.186

The lecture identifies **ICANN** as the organization associated with governance/control of Internet names and numbers.

12. ICANN ★ ★ ★

Full Form

ICANN = Internet Corporation for Assigned Names and Numbers

The lecture describes ICANN as a **non-profit, multi-stakeholder organization** with important roles in Internet names and numbers.

Major Roles

1. Internet Names

ICANN is involved in the assignment/coordination of Internet domain names.

2. Internet Addresses

ICANN is involved in Internet address allocation/coordination.

3. Name-related Policies

It also has policies for handling certain domain-name disputes.

13. DNS and Name → IP Mapping

The lecture mentions the mapping:

gaia.cs.umass.edu

↓

DNS

↓

128.119.40.186

Important distinction

ICANN → governance/coordination of Internet names and numbers.

DNS → performs the actual **domain-name resolution** used to translate names into IP addresses.

★ GATE Trap

✗ ICANN directly performs every DNS lookup.

☑ DNS performs name resolution; ICANN has governance/coordination roles related to names and numbers.

14. ICANN's Multi-Stakeholder Approach

The lecture describes ICANN as using a:

Bottom-up, consensus-based, multi-stakeholder process

Participants include:

- Governments
- Businesses
- Academia
- Civil society

15. Layer 3 — Internet Content ★ ★

The highest layer concerns:

- What content is available?
- Who can access it?
- How content is regulated/filtered?

This layer has strong:

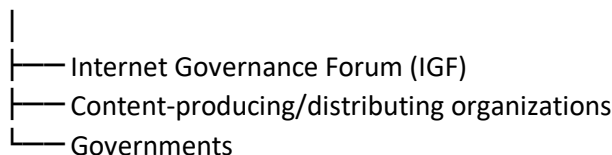
- Social
- Policy
- Economic
- Governance

dimensions.

Main Actors

The lecture identifies three important groups:

Internet Content



16. Internet Governance Forum (IGF)

IGF

Internet Governance Forum

It was convened for **multi-stakeholder discussions** about Internet policy and governance.

Important

IGF is primarily a **forum for discussion**, not a decision-making body in the same sense as organizations/governments that make binding decisions.

Its role is to:

- Inform
- Inspire
- Facilitate policy discussion

★ GATE Trap

IGF = **decision-making body**

✗ False according to the lecture.

IGF = **discussion/forum for Internet governance and policy.**

17. Governments and Internet Content

Governments can establish laws and regulations within their jurisdictions.

They may influence:

- Internet access
- Internet content
- Content regulation
- Filtering/blocking

The lecture identifies governments as one of the major actors at the content layer.

18. Firewall for Content/Access Control ★ ★ ★

A **firewall** can be configured not only to forward traffic but also to **block traffic according to specified rules**.

Example:

User



Access Network



Firewall



Allowed → Forward



Blocked → Drop



Content Server

A firewall may have a rule such as:

IF destination IP = X

THEN DROP packet

The lecture gives an example of blocking packets destined for a particular IP address.

19. Deep Packet Inspection (DPI) ★ ★ ★

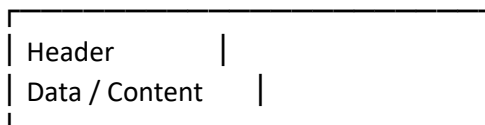
Some firewalls can perform **Deep Packet Inspection**.

DPI

The firewall examines information **inside a packet** and can make forwarding/blocking decisions based on packet contents.

Example:

Packet



Firewall



Inspect content



Forward / Drop

Example rule:

IF packet contains "X"

THEN DROP

GATE Concept

Normal routing decision → generally based on packet headers.

DPI → can inspect deeper/inside packet content for filtering decisions.



Important Organizations — One Table

Organization **Main Role**

IETF Internet architecture & protocol standards

RFC Documents/specifications for Internet standards

3GPP	3G/4G/5G mobile standards
IEEE 802.3	Ethernet
IEEE 802.11	Wi-Fi
ITU	International telecommunications standards
ICANN	Internet names & numbers governance/coordination
IGF	Multi-stakeholder Internet governance discussion

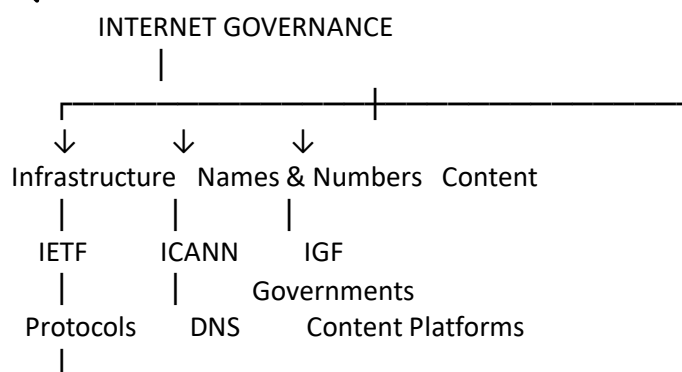
★ Three-Layer Revision Table

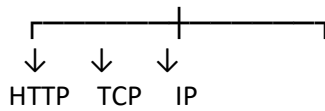
Layer	Main Concern	Examples/Actors
Infrastructure	Technical operation	Protocols, routers, switches, formats; IETF
Names & Numbers	Names/IP addresses	ICANN, DNS
Content	Content & access	IGF, content organizations, governments, firewalls

⚠ GATE Traps — Must Remember

- Internet is controlled by one organization.**
✗ False — governance involves multiple stakeholders.
- IETF mainly governs domain-name ownership.**
✗ False — IETF focuses on Internet architecture and technical standards.
- 3GPP → 3G/4G/5G mobile standards.**
✓ Correct.
- 802.3 → Wi-Fi.**
✗ False.
802.3 → Ethernet
- 802.11 → Wi-Fi.**
✓ Correct.
- ICANN performs DNS lookup for every user.**
✗ False.
DNS performs name resolution.
- IGF is primarily a decision-making body.**
✗ False — it is a forum for discussion.
- Firewall can only forward packets.**
✗ False — it can apply blocking rules.
- DPI only checks destination IP address.**
✗ False — DPI can inspect packet contents.
- Standards are unnecessary if networks are interconnected.**
✗ False — standards enable interoperability.

🗺 30-Second GATE Revision





3GPP → 3G/4G/5G

802.3 → Ethernet

802.11 → Wi-Fi

Firewall → Blocking

DPI → Inspect packet contents

🎯 For GATE, learn these 8 things perfectly:

1. **Three layers:** Infrastructure → Names/Numbers → Content

2. **IETF → Internet technical standards**

3. **RFC → Internet technical specifications/documents**

4. **3GPP → 3G/4G/5G**

5. **802.3 → Ethernet; 802.11 → Wi-Fi**

6. **ICANN → Internet names & numbers**

7. **IGF → governance/policy discussion forum**

8. **Firewall + DPI → traffic/content-based access control**

These are the parts of the transcript most worth putting into your **GATE revision notebook**; the historical stories and lecture anecdotes can be skipped.

Introduction to the Internet & Protocols

1. Internet

The **Internet** is a global interconnected system of **networks, hosts, packet switches, and communication links** that enables data communication.

Two views of the Internet

1. Nuts-and-Bolts View

- Focuses on physical and logical components of the Internet.
- Main components:
 - **Hosts / End Systems**
 - **Routers**
 - **Switches**
 - **Communication Links**
 - **Networks**

2. Service View

- The Internet acts as a **platform for applications**.
- Applications use the network to **send and receive data**.
- Example: Web browsing, video streaming, email, etc.

2. Hosts / End Systems

Host (End System): A device located at the **edge of the network** that communicates over the Internet.

Examples

- Computer/Laptop
- Smartphone
- Server
- IoT devices
- Smart appliances
- Connected vehicles

GATE Point:

Hosts are the **end points** of communication.

3. Packet Switches

A **packet switch** is a network device that forwards **packets of data** from one link to another.

Two important types:

Router

- Typically forwards packets **between different networks**.

Switch

- Typically forwards packets **within a network**, especially a LAN.
Both routers and switches perform packet forwarding, but they operate at different networking scopes/layers.

4. Communication Links

Communication links connect:

- Hosts
- Routers
- Switches
- Other network devices

They provide the path through which data is transmitted.

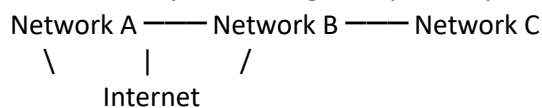
Examples include:

- Wired links
- Wireless links
- Fiber-optic links

5. Internet = Network of Networks ★

The Internet is **not a single network**.

It is formed by connecting multiple independently operated networks.



Hence:

The Internet is a network of networks.

6. Protocol

A **protocol** is a set of rules that governs communication between network entities.

Network entities may include:

- Hosts
- Routers
- Switches
- Applications

Formal Definition ★

A protocol defines the **format and order of messages** exchanged among network entities and the **actions taken on message transmission and/or receipt**.

Three Important Components

Protocol defines Meaning

Format Structure/content of messages

Order Sequence in which messages are exchanged

Actions What an entity does after sending/receiving a message

Example

Client Server
| ---- Request -----> |
| <---- Response ----- |

This is a simple **request-response** communication pattern.

7. Human Protocol → Network Protocol

Human communication also follows protocols.

Example:

Person A → "What time is it?"

Person B → Checks watch

Person B → Gives the time

Similarly, computer network entities exchange messages according to predefined rules.

Therefore:

Network protocols are analogous to human communication protocols, but are executed by network entities.

8. Protocol Phases

A communication protocol may consist of multiple phases.

A simple example:

Connection / Greeting

↓

Request

↓

Response

This concept is also used in many Internet protocols.

9. Internet Standards

IETF

IETF = Internet Engineering Task Force

It is an organization involved in developing and maintaining **Internet standards and protocols**.

RFC

RFC = Request for Comments

Internet protocol specifications and standards are published as **RFC documents**.

Remember

IETF

↓

Internet Standards / Protocols

↓

RFC Documents

★ GATE Quick Revision

Internet

Network of Networks

Host

End system connected to the network

Packet Switch

Forwards packets

Types:

- Router
- Switch

Protocol

Format + Order + Actions

IETF

Internet Engineering Task Force

RFC

Request for Comments



Most Important GATE Points

1. **Internet = Network of Networks**
2. **Hosts/End Systems are at the network edge.**
3. **Routers and switches are packet switches.**
4. **Protocols govern communication between network entities.**
5. A protocol specifies:
 - **Message format**
 - **Message order**
 - **Actions on transmission/reception**
6. **IETF** is associated with Internet standards.
7. **RFCs** document Internet protocols and standards.
8. Internet can be viewed as both:
 - **Infrastructure (nuts-and-bolts view)**
 - **Service platform for applications**

~Anshu